

Homework 1: Due on March 15

Caution: Only **hand-written solutions** are allowed, and typed solutions are **not** accepted. When solving homework problems, it is important to show your derivation at each step. Nowadays, many online tools make it easy to find answers, but the primary goal of these assignments is to deepen your understanding through hands-on problem-solving. By working through the calculations yourself, you engage more deeply with the material, making the learning process more meaningful, rather than simply copying answers from external sources.

1 Euler Characteristic

1. Find the Euler characteristic of a Klein bottle (Figure 1).
2. Find the area of the following triangle (Figure 3) bounded by great circles on a 2-sphere with radius one.
3. Generalize it to the area of n -gon on a 2-sphere with radius one. (An example of a pentagon is drawn in Figure 4.)
4. Prove the Euler characteristic of a 2-sphere is equal to two by using a cell decomposition (Figure 2).

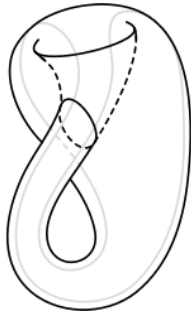


Figure 1: A Klein bottle

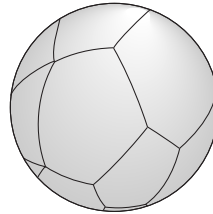


Figure 2: cell decomposition of a 2-sphere

2 Real projective spaces

For elements $x = (x^0, \dots, x^n)$ and $y = (y^0, \dots, y^n)$ of $\mathbb{R}^{n+1} \setminus \{0\}$, we define an equivalence relation $x \sim y$ by

$$x = \alpha y \quad \forall \alpha \in \mathbb{R}.$$

Let us define $\mathbb{RP}^n$ by $(\mathbb{R}^{n+1} \setminus \{0\}) / \sim$. Show that $\mathbb{RP}^n$ is a manifold and $\mathbb{RP}^1$ is diffeomorphic to S^1 . The space is called a real projective space.

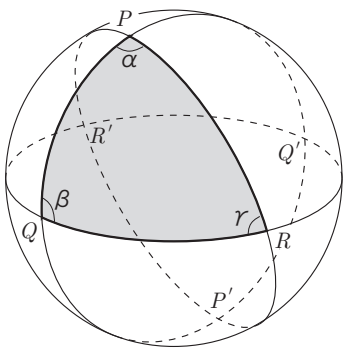


Figure 3: A triangle on a 2-sphere

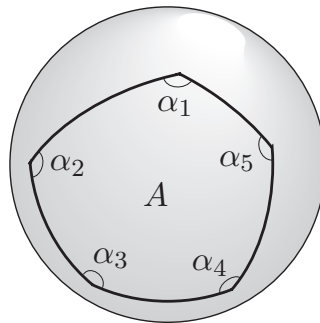


Figure 4: A pentagon on a 2-sphere

3 Vector fields

We construct vector fields on an n -sphere $S^n \subset \mathbb{R}^{n+1}$ in Example 2.11 of the lecture note.

$$Y = \sum Y^i \frac{\partial}{\partial x^i}$$

where

$$Y^i = \begin{cases} (-x^1, x^0, -x^3, x^2, \dots, -x^{2k+1}, x^{2k}) & n = 2k + 1 \\ (-x^1, x^0, -x^3, x^2, \dots, -x^{2k+1}, x^{2k}, 0) & n = 2k + 2 \end{cases}.$$

For $n = 1, 2, 3$, write explicitly the flows generated by this vector field.